

laboratory experiments that have formed key pillars of the field, such as those experiments depicted in Examples 1 and 2 in the book's Introduction section. The thought experiments in Section 1 are, for the most part, re-castings of the simple cognitive tests devised by psychologists and economists over the past three-to-four decades to illustrate the fallacies, miscalculations, and biases distinguishing *Homo sapiens* from *Homo economicus*. Similarly, the laboratory experiments presented in Section 2 are, for the most part, re-castings of the seminal experiments conducted by Kahneman and Tversky (among many others). These experiments helped motivate the revised theories of human choice behavior, such as Kahneman and Tversky's (1979) Prospect Theory, which form another pillar of behavioral economics. Alongside these experiments, Section 2 presents the revised theories of human choice behavior with varying degrees of rigor. This is where the theoretical bases of *Homo economicus*' rational choice behavior are examined, and where key refinements to this theory are developed—theoretical refinements underpinning the myriad departures from rational choice behavior we witness *Homo sapiens* make in this section's laboratory and field experiments (and which are examined further in Sections 3 and 4).

Section 3 submerses the student in the world of behavioral game theory. Here we explore games such as Ultimatum Bargaining presented in Example 5. We follow Camerer (2003)'s lead, first by characterizing the games analytically (i.e., identifying solution, or equilibrium, concepts that are predicted to result when members of *Homo economicus* play the games), and then by discussing empirical results obtained from corresponding field experiments conducted with *Homo sapiens*. It is within the context of these games and field experiments that theories of social interaction are tested concerning *inter alia* trust and trustworthiness, honesty, fairness, reciprocity, etc. As with the thought and laboratory experiments presented in Sections 1 and 2, the games and field experiments presented in Section 3 are meant to be replicated with students as subjects and the instructor as the experimenter, or researcher.

Finally, Section 4 wades into the vast sea of empirical research and choice architecture. Here the student explores studies reporting on (1) the outcomes of actual policy nudges, such as the SMarT retirement-savings plan presented in Example 3 of the Introduction, (2) analyses of secondary datasets to test for choice behavior consistent with the revised theories discussed in Section 2, such as the test for loss aversion in Example 4 of the Introduction, and (3) analyses of primary datasets obtained from novel field experiments to further test the revised theories. The main purpose of this section is not only to introduce the student to interesting empirical studies and policy adaptations in the field of behavioral economics, but also, in the process, to incubate in the student an abiding appreciation for the obscure settings that sometimes lend themselves to such study.<sup>3</sup>

## THE TEXTBOOK'S DIFFERENT LEVELS OF RIGOR

Because the mathematical and computational rigor of material presented in this textbook varies throughout, particularly in Sections 2 – 4, the extent of the rigor used in the presentation of a given topic is indicated with superscripts. Topics without a superscript are considered basic and universal enough that backgrounds in economics, mathematics, or statistics are not required for the reader to understand the material. Topics with a single asterisk (\*) indicate that higher mathematical reasoning skills are recommended for the reader to fully grasp the material. Topics with a double

3. Our approach to studying behavioral economics is focused on the underlying laboratory experimentation and behavioral games that form the bedrock of the field. As such, we eschew delving into related fields such as neuroeconomics and auction theory. See Cartwright (2018) and Just (2013) for introductions to the former and latter fields, respectively.